

Year 11

Mathematics

Yearly Examination

2014

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- In Questions 11- 14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II

60 marks

- Attempt Questions 11-14
- Start each question in a new writing booklet
- Write your name on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your name and “N/A” on the front cover
- Allow about 1 hour 45 minutes for this section

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Section I

10 Marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1 – 10.

1 What is the midpoint of the points (3, 8) and (–5, –2)?

(A) (–1, 3)

(B) (4, 5)

(C) (–2, 6)

(D) (2, 10)

2 Solve $\cos \theta = -\frac{\sqrt{3}}{2}, 0 \leq \theta \leq 360^\circ$.

(A) $\theta = 210^\circ, 330^\circ$

(B) $\theta = 150^\circ, 330^\circ$

(C) $\theta = 150^\circ, 210^\circ$

(D) $\theta = 150^\circ, 210^\circ, 330^\circ$

3 The solutions to $|x+1| = 4$ are

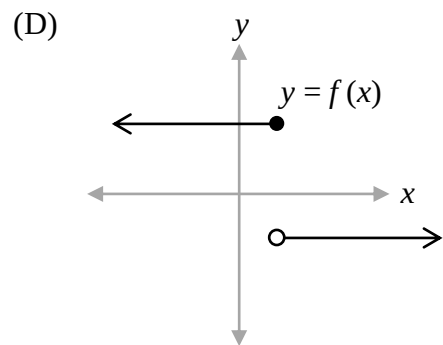
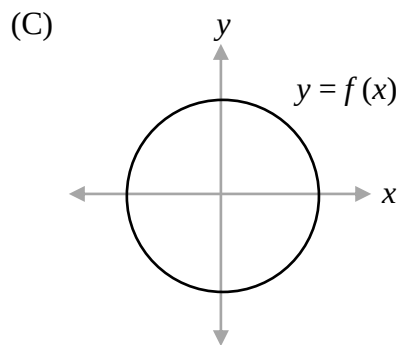
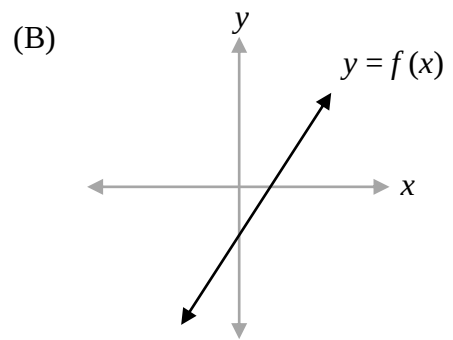
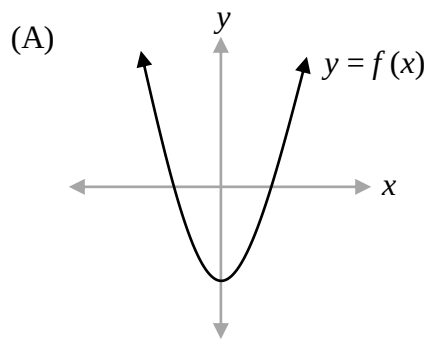
(A) $x = 4$

(B) $x = 3$ and $x = -5$

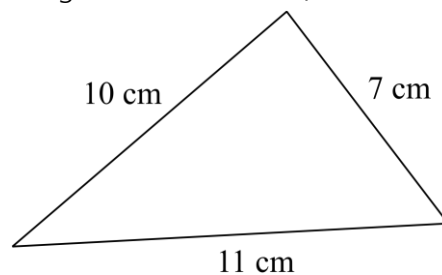
(C) $x = 5$ and $x = -4$

(D) $x = -5$

4 Which of the following graphs is NOT a function?



5 The following triangle has sides 7 cm, 10 cm and 11 cm.



Not to scale

Let A is the smallest angle.

Which of the following expressions is correct for angle A ?

(A) $\cos A = \frac{7^2 + 11^2 - 10^2}{2 \times 7 \times 11}$

(B) $\cos A = \frac{10^2 + 7^2 - 11^2}{2 \times 7 \times 10}$

(C) $\cos A = \frac{10^2 + 11^2 - 7^2}{2 \times 10 \times 11}$

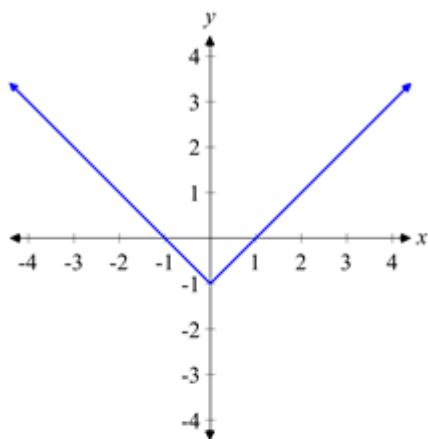
(D) $\cos A = \frac{10^2 + 7^2 - 11^2}{2 \times 10 \times 11}$

6 Simplify $\frac{\cos(90^\circ - \theta)}{\sin(90^\circ - \theta)}$:

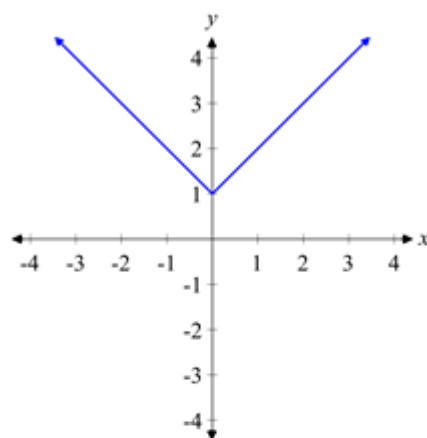
- (A) 1
- (B) $\cot\theta$
- (C) $\tan\theta$
- (D) $-\tan\theta$

7 Which graph best represents $y = |x| - 1$?

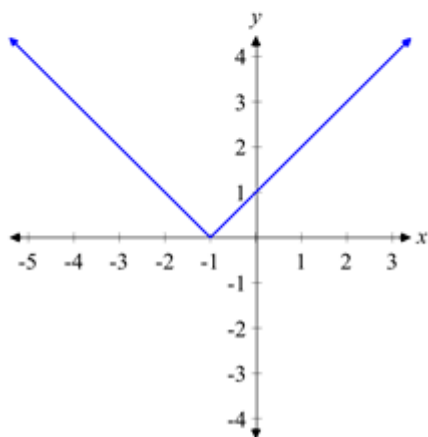
(A)



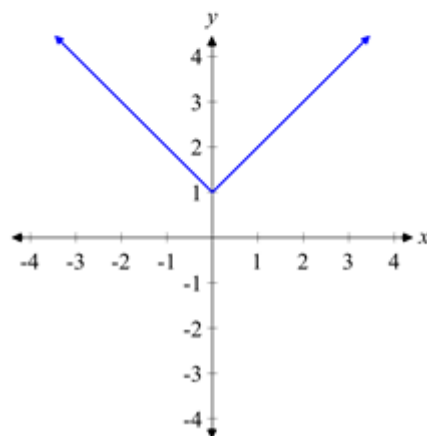
(B)



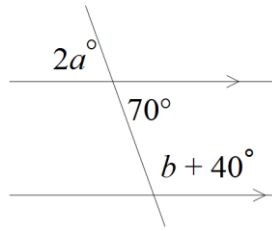
(C)



(D)



- 8 What are the values of a and b ?



- (A) $a = 35^\circ, b = 30^\circ$
- (B) $a = 35^\circ, b = 70^\circ$
- (C) $a = 70^\circ, b = 30^\circ$
- (D) $a = 70^\circ, b = 70^\circ$
- 9 Two points A and B lie on opposite sides of the line l . A and B are equidistant from the line l . Which statement is always true?
- (A) The line through A and B is perpendicular to the line l .
- (B) The line through A and B is parallel to the line l .
- (C) The midpoint of AB lies on the line l .
- (D) If a line passing through A is parallel to a line passing through B , both these lines are parallel to the line l .
- 10 The line $6x - ky = 2$ passes through the point $(3, 2)$. What is the value of k ?
- (A) $\frac{10}{3}$
- (B) $-\frac{10}{3}$
- (C) -8
- (D) 8

END OF SECTION I

Section II

60 marks

Attempt Questions 11 - 14

Allow about 1 hour 45 minutes for this section

Answer each question in a SEPARATE writing booklet.

All necessary working should be shown in every question

Marks

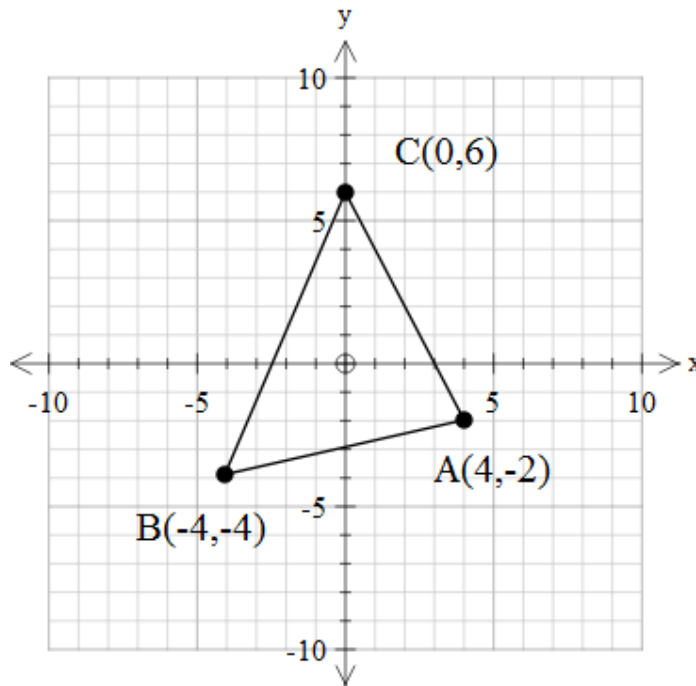
Question 11 (15 marks). Use a SEPARATE writing booklet.

- | | | |
|----|---|---|
| a) | Factorise completely: | |
| | (i) $a^2 - 4a + 4$ | 1 |
| | (ii) $ab - 2bc + 5a - 10c$ | 2 |
| b) | Solve: | |
| | (i) $8(x + 2) = 9 - x$ | 2 |
| | (ii) $\frac{5y}{2} - \frac{2y}{3} = 3$ | 2 |
| c) | Expand and simplify $3(4x - 7)^2 - (12x - 8x^2)$. | 2 |
| d) | Simplify $\frac{4k^3 \times 6k^4}{12k^5}$ | 2 |
| e) | State the domain and range of $y = \sqrt{x - 5}$ | 2 |
| f) | Solve these simultaneous equations.
$2x + y = 1$
$6x - 4y = 10$ | 2 |

END OF QUESTION 11

Marks**Question 12** (15 marks). Use a SEPARATE writing booklet.

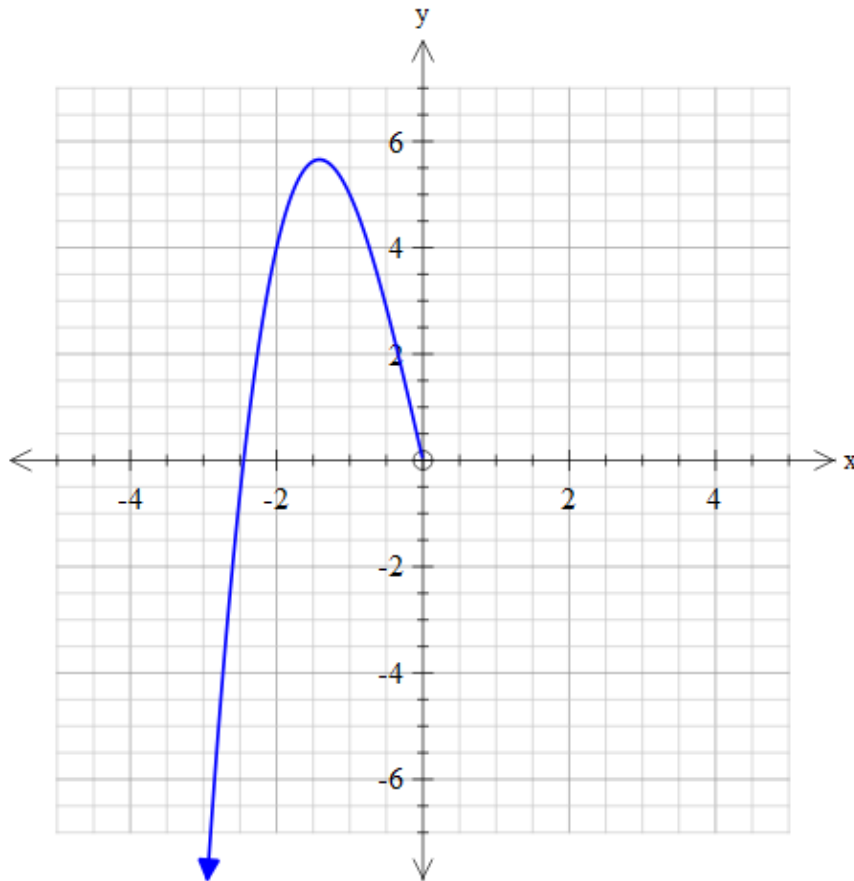
- a) $A(4, -2), B(-4, -4), C(0, 6)$ are the vertices of a triangle ABC



- | | |
|---|---|
| (i) Find the midpoint of AC. | 1 |
| (ii) Find the exact length of BC. | 1 |
| (iii) Show the equation of the line BC is $5x - 2y + 12 = 0$ | 2 |
| (iv) Calculate the exact perpendicular distance of A from the line BC | 2 |
| (v) Find the area of the triangle ABC. | 1 |
| (vi) Find the coordinates of D such that ABCD is a parallelogram. | 2 |

QUESTION 12 CONTINUES ON THE NEXT PAGE

- b) (i) Show that $f(x) = x^3 - 6x$ is an odd function 2
- (ii) Below is an incompleted graph of the function $f(x) = x^3 - 6x$. 1
 Copy or trace the diagram into your writing booklet
 and then **complete** the graph.



- c) Find the equation of the **normal** to the curve $f(x) = x^2 - 4x + 1$ at the point where $x = 1$. 3

END OF QUESTION 12

Marks**Question 13** (15 marks). Use a SEPARATE writing booklet.

- a) Differentiate with respect to x and fully simplify your answer where possible:
- | | |
|------------------------------|---|
| (i) $3x^4 + 7x^2 + 1$ | 1 |
| (ii) $\sqrt{x^2 + 3}$ | 2 |
| (iii) $\frac{2}{\sqrt{x}}$ | 2 |
| (iv) $\frac{3x + 1}{3 - 4x}$ | 2 |
| (v) $(2x - 1)(x - 2)^4$ | 2 |
- b) Given the curve $f(x) = x^2 - 4x$
- | | |
|--|---|
| (i) Show that $f(x + h) = x^2 - 2xh + h^2 - 4x - 4h$ | 1 |
| (ii) Hence, differentiate $f(x) = x^2 - 4x$ from first principles. | 2 |
- c) The function $y = f(x)$ is defined as follows:
- $$f(x) = \begin{cases} 7 & \text{for } x < 0 \\ x^2 + 3 & \text{for } x \geq 0 \end{cases}$$
- | | |
|---------------------------------------|---|
| (i) Find the value of $f(-1)$. | 1 |
| (ii) Sketch the function $y = f(x)$. | 2 |

END OF QUESTION 13

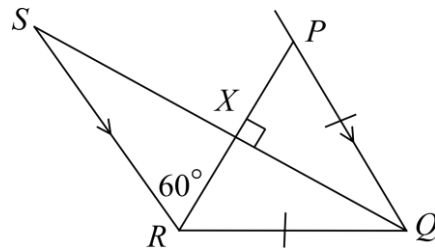
Marks**Question 14** (15 marks). Use a SEPARATE writing booklet.

- a) Sketch the graph and clearly label the vertex, asymptotes (if any) and the points where the graph cuts the x and y -axes.

(i) $y = x^2 - x - 6$ 3

(ii) $y = \frac{4}{x-2}$ 3

- b) In the diagram below, PQ is parallel to SR , $PQ = QR$, $\angle XRS = 60^\circ$ and lines PR and SQ are perpendicular.



Copy or trace the diagram into your writing booklet

- (i) Find the size of $\angle RQS$. Give reasons. 2
- (ii) Prove that $\triangle QRS$ is an isosceles triangle. 2
- c) Prove $\sin \theta \cos \theta \tan \theta = 1 - \cos^2 \theta$ 2
- d) Draw a sketch showing the region on the number plane where the inequalities $x^2 + y^2 \leq 4$ and $x > 1$ hold simultaneously. 3

END OF EXAMINATION

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2014 Year 11 Yearly Examination

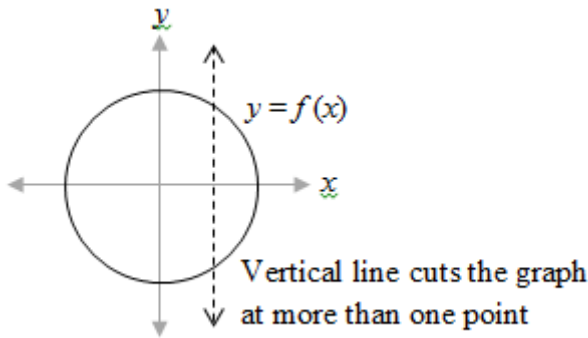
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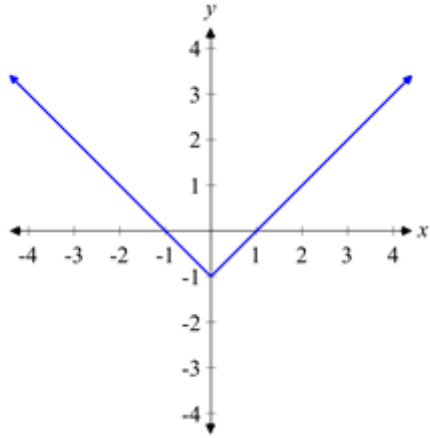
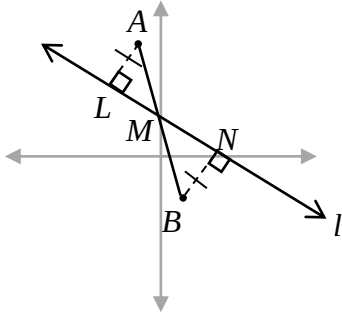
Preliminary Mathematics

Section A Multiple-Choice Answer Sheet

- | | | | | |
|-----------|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 | A ☐ | B ☐ | C ☐ | D ☐ |
| 2 | A ☐ | B ☐ | C ☐ | D ☐ |
| 3 | A ☐ | B ☐ | C ☐ | D ☐ |
| 4 | A ☐ | B ☐ | C ☐ | D ☐ |
| 5 | A ☐ | B ☐ | C ☐ | D ☐ |
| 6 | A ☐ | B ☐ | C ☐ | D ☐ |
| 7 | A ☐ | B ☐ | C ☐ | D ☐ |
| 8 | A ☐ | B ☐ | C ☐ | D ☐ |
| 9 | A ☐ | B ☐ | C ☐ | D ☐ |
| 10 | A ☐ | B ☐ | C ☐ | D ☐ |

SECTION I

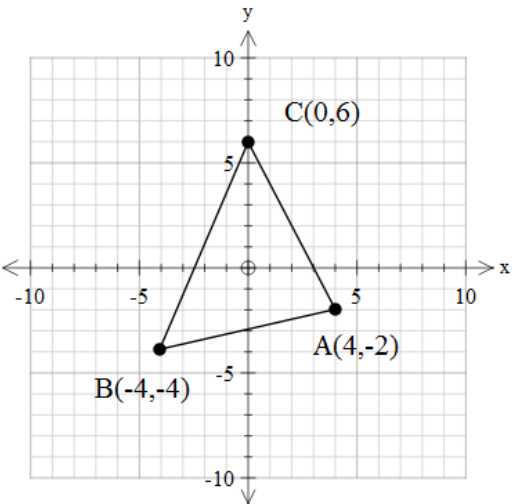
1	$(x, y) = \left(\frac{3 + (-5)}{2}, \frac{8 + (-2)}{2} \right)$ $= (-1, 3)$	A
2	$\cos \theta = -\frac{\sqrt{3}}{2}$ $\cos 30^\circ = \frac{\sqrt{3}}{2}$ $\cos$ is negative in the 2nd and 3rd quadrants. $\therefore \theta = 180^\circ - 30^\circ, 180^\circ + 30^\circ$ $= 150^\circ, 210^\circ$	C
3	$ x + 1 = 4 \therefore x + 1 = 4 \therefore x = 3$ and $x + 1 = -4 \therefore x = -5$	B
4	 Vertical line cuts the graph at more than one point	C
5	Smallest angle is opposite the smallest side. $\cos A = \frac{10^2 + 11^2 - 7^2}{2 \times 10 \times 11}$	C
6	$\frac{\cos(90^\circ - \theta)}{\sin(90^\circ - \theta)} = \frac{\sin \theta}{\cos \theta} = \tan \theta$	C

7		A
8	$2a = 70, a = 35$ $70 + b + 40 = 180, b = 70$	B
9	$\triangle ALM \equiv \triangle BNM$ (AAS) $\therefore AM = BM$ (corresponding sides of congruent triangles) $\therefore M$ is the midpoint of AB and it lies on the line l. 	C
10	$6x - ky = 2, (3, 2)$ $6 \times 3 - 2k = 2$ $-2k = -16$ $k = 8$	D

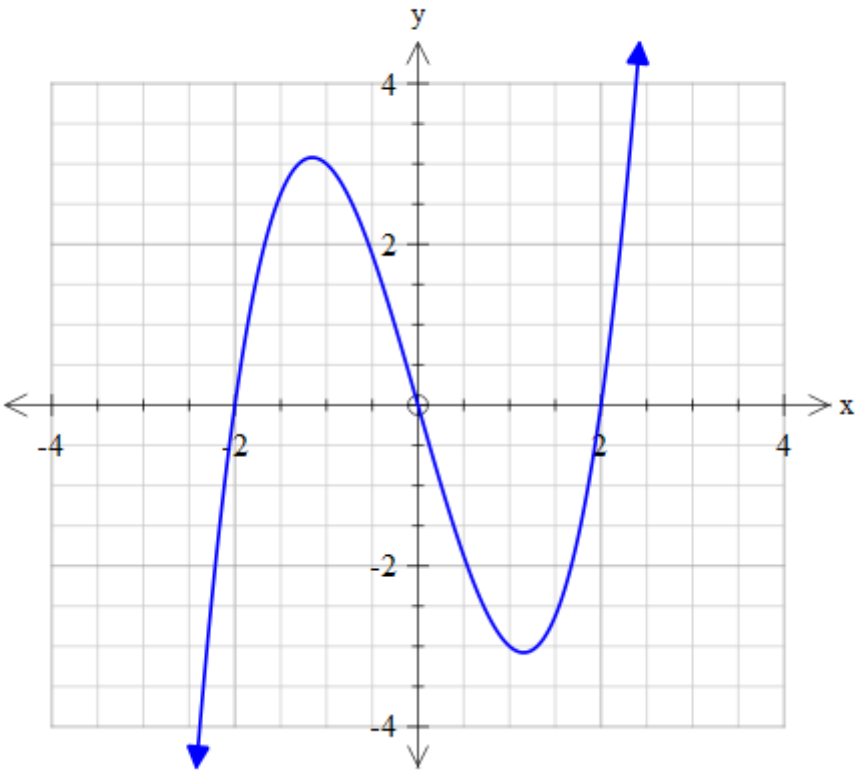
END OF SECTION I

SECTION II

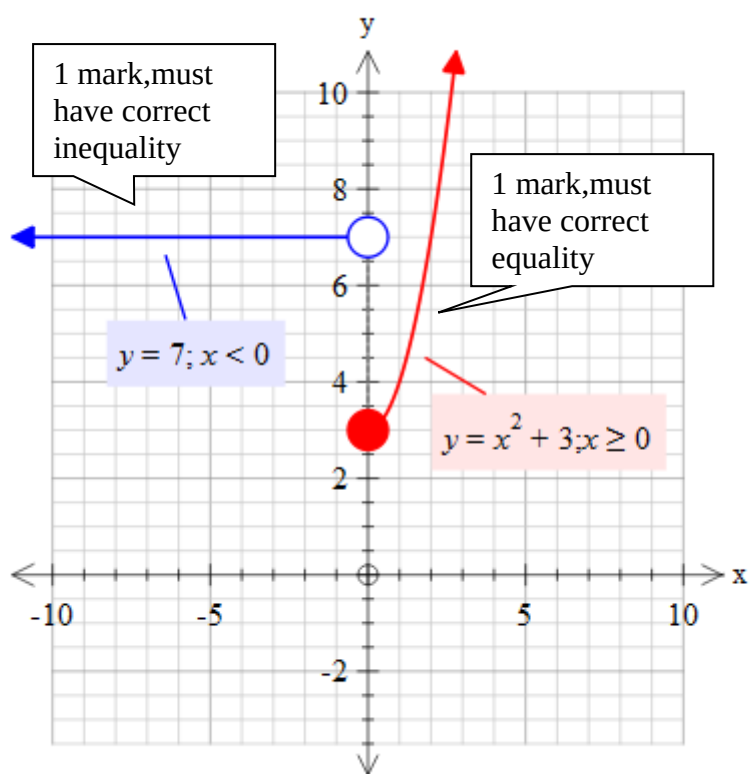
11		
a		
i	$a^2 - 4a + 4 = (a - 2)^2$	1
ii	$ab - 2bc + 5a - 10c = b(a - 2c) + 5(a - 2c)$ $= (a - 2c)(b + 5)$	1 mark 1 mark
b		
i	$8(x + 2) = 9 - x$ $8x + 16 = 9 - x$ $9x = -7$ $x = \frac{-7}{9}$	1 mark (correct expansion) 1 mark
ii	$6 \times \left(\frac{5y}{2} - \frac{2y}{3} \right) = 3 \times 6$ $15y - 4y = 18$ $11y = 18$ $y = \frac{18}{11} = 1\frac{7}{11}$	1 mark 1 mark
c	$\frac{4k^3 \times 6k^4}{12k^5}$ $= \frac{24k^7}{12k^5}$ $= 2k^2$	1 mark 1 mark
d	Domain: $x \geq 5$ Range: $y \geq 0$	1 mark each

e	$\begin{cases} 2x + y = 1 & (1) \times 3 \\ 6x - 4y = 10 & (2) \end{cases}$ $\begin{cases} 6x + 3y = 3 \\ 6x - 4y = 10 \end{cases}$ $7y = -7$ $y = -1$ Sub $y = -1$ into (1) $2x - 1 = 1$ $x = 1$ $\therefore x = 1; y = -1$	2 marks correct values for x and y
f	$f(x) = x^2 - 4x + 1$ At $x = 1$ $f(1) = 1^2 - 4 \times 1 + 1 = -2$ $f'(x) = 2x - 4$ At $x = 1$ $f'(1) = 2 \times 1 - 4 = -2$ ie gradient of tangent is -2 Equation of the tangent at (1, -2) $y = -2x + b$ sub(1, -2) $-2 = -2 + b$ $b = 0$ $y = -2x \text{ or } 2x + y = 0$	1 mark gradient of tangent -2 1 mark correct line
12		
a		
		
i	$mid_{AC} = \left(\frac{4+0}{2}; \frac{-2+6}{2} \right) = (2, 2)$	1
ii	$d = \sqrt{(-4-0)^2 + (-4-6)^2}$ $= \sqrt{16+100} = \sqrt{116} = 2\sqrt{29}$	1

iii	$m = \frac{-4-6}{-4-0} = \frac{5}{2}$ $y-6 = \frac{5}{2}(x-0)$ $y-6 = \frac{5x}{2}$ $2y-12 = 5x$ $0 = 5x - 2y + 12$	1 mark gradient 1 mark line
iv	$d = \frac{ Ax_1 + By_1 + C_1 }{\sqrt{A^2 + B^2}}$ $= \frac{ 5 \times 4 - 2 \times -2 + 12 }{\sqrt{25 + 4}}$ $= \frac{36}{\sqrt{29}}$	1 mark correct substitution correct formula 1 mark
v	$A = \frac{1}{2} \times 2\sqrt{29} \times \frac{36}{\sqrt{29}} = 36u^2$	1
vi	Midpoint CA is (2,2) $2 = \frac{x_D - 4}{2} \rightarrow x_D = 8$ Therefore $2 = \frac{y_D - 4}{2} \rightarrow y_D = 8$ $D = (8,8)$ Other correct methods accepted. Accept point in Q2 (-8,4) or (0,-12)	2 marks each correct coordinate OR 1 mark correct progress
b		
i	$f(x) = x^3 - 4x$ $f(-x) = (-x)^3 - 4(-x)$ $= -x^3 + 4x$ $= -(x^3 - 4x)$ $= -f(x)$ Function is odd since $f(-x) = -f(x)$	1 mark explicit sub of -x. 1 mark arriving at correct result

ii		1 mark correct x intercept x=2 1 mark correct shape
c	$y = 2x^3 + ax + b$ At (1, 1) $1 = 2 + a + b$ $a + b = -1 \quad (1)$ $y' = 6x + a$ $x = 1, y' = 6 + a$ Parallel to the line $y = x, y' = 1$ so $6 + a = 1, a = -5$ Sub $a = -5$ into (1) $b = 4$ $\therefore a = -5, b = 4$	1 mark correct progress to equating $y' = 1$ 1 mark correct values of a and b
13		
a		
i	$\frac{d}{dx}(3x^4 + 7x^2 + 1) = 12x^3 + 14x$ 1 mark	1
ii	$\frac{d}{dx}\sqrt{x^2 + 3} = \frac{d}{dx}(x^2 + 3)^{\frac{1}{2}} = \frac{1}{2}(x^2 + 3)^{-\frac{1}{2}} \times 2x = \frac{x}{\sqrt{x^2 + 3}}$ 1 mark	2
iii	$\frac{d}{dx} \frac{2}{\sqrt{x}} = \frac{d}{dx} 2x^{-\frac{1}{2}} = 2 \cdot \frac{-1}{2} x^{-\frac{3}{2}} = -\frac{1}{x\sqrt{x}}$ 1 mark 1 mark	2

iv	$\frac{d}{dx} \left(\frac{3x+1}{3-4x} \right) = \frac{3(3-4x) + 4(3x+1)}{(3-4x)^2}$ $= \frac{9-12x+12x+4}{(3-4x)^2}$ $= \frac{13}{(3-4x)^2}$	1 mark correct use of quotient rule 1 mark correct
v	$\frac{d}{dx} (2x-1)(x-2)^4 = 2(x-2)^4 + 4(x-2)^3(2x-1)$ $= 2(x-2)^3[(x-2) + 2(2x-1)]$ $= 2(x-2)^3(5x-4)$	1 mark correct use of product rule 1 mark correct
b		
i	$f(x) = x^2 - 4x$ $f(x+h) = (x+h)^2 - 4(x+h)$ $= x^2 + 2xh + h^2 - 4x - 4h$	1
ii	$f(x) = x^2 - 4x$ $f'(x) = \lim_{h \rightarrow 0} \frac{[(x+h)^2 - 4(x+h)] - (x^2 - 4x)}{h}$ $= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 4x - 4h - x^2 + 4x}{h}$ $= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 4h}{h}$ $= \lim_{h \rightarrow 0} \frac{h(2x + h - 4)}{h}$ $= \lim_{h \rightarrow 0} 2x + h - 4$ $= 2x - 4$	1 mark NOTE lim must $\lim_{h \rightarrow 0}$ appear in every line until last 1 mark
c		
i	$f(-1) = 7$	1



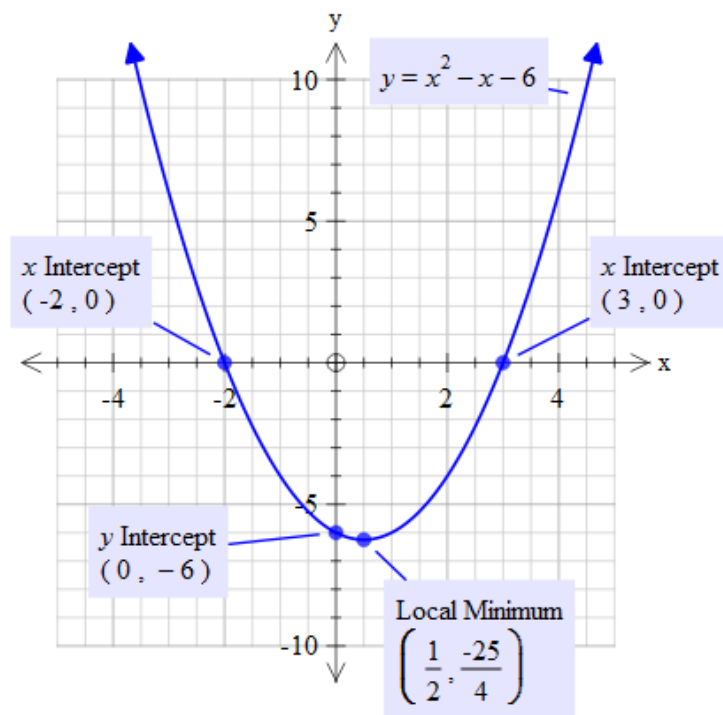
ii

2

14

a

i

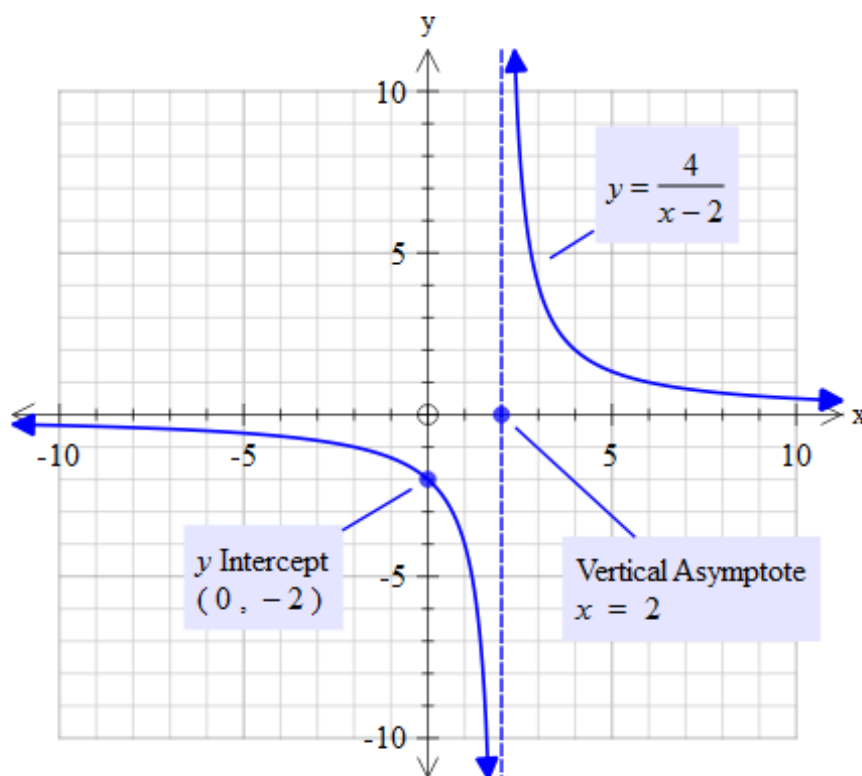


1 mark all
intercepts
correct

1 mark cords
of vertex
correct

1 mark
correct shape

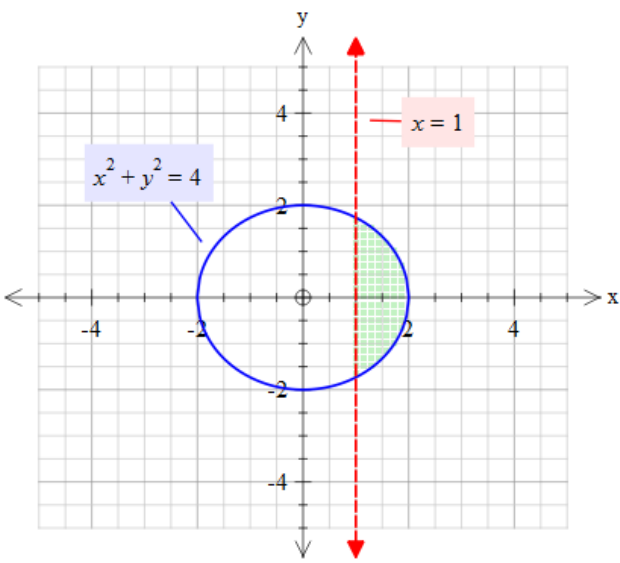
ii



1 mark
correct VA at
 $x=2$

1 mark
correct y
intercept

1 mark
correct shape

b		
i	$\angle QPX = \angle XRS = 60^\circ$ (alternate angles are equal, $PQ \parallel SR$) ΔPQR is an isosceles triangle (two equal sides, $PQ = QR$) $\angle QPX = \angle QRX = 60^\circ$ (base angles are equal, isosceles triangle) $\angle RXQ + \angle PXQ = 180^\circ$ (adjacent angles on a straight line) $\angle RXQ + 90^\circ = 180^\circ$ $\angle RXQ = 90^\circ$ In ΔQXR $\angle RQX + \angle RXQ + \angle QRX = 180^\circ$ $\angle RQX + 90^\circ + 60^\circ = 180^\circ$ $\angle RQX = 30^\circ$ or $\angle RQS = 30^\circ$	2.
ii	In ΔQRS $\angle QSR + \angle QRS + \angle RQS = 180^\circ$ (Angle sum of a triangle is 180°) $\angle QSR + (60^\circ + 60^\circ) + 30^\circ = 180^\circ$ $\angle QSR = 30^\circ$ $\therefore \angle RQS = \angle QSR = 30^\circ$ ΔQRS is an isosceles triangle (two base angles are equal).	2
c	LHS = $\sin \theta \cos \theta \tan \theta$ $= \sin \theta \cos \theta \times \frac{\sin \theta}{\cos \theta}$ $= \sin^2 \theta$ $= 1 - \cos^2 \theta$ $= \text{RHS}$	1 mark for each correct substitution
d	$x^2 + y^2 = 4$ is a circle centre (0,0) and radius 2 $x = 1$ is a vertical line passing through (1,0) 	1 mark correct circle 1 mark correct $x > 1$ 1 mark correct region